

2.4.2 Intermediate Value Theorem Worksheet

1. Show precisely how the *Intermediate Value Theorem* is used to prove that the equation $x^3 - 2x - 3 = 0$ has at least one real solution.

Let $f(x) = x^3 - 2x - 3$. We want to know if $f(x)$ has a root.

f is continuous because f is a polynomial.

Note that:

$$f(1) = 1 - 2 - 3 = -4 < 0$$

$$f(2) = 8 - 4 - 3 = 1 > 0$$

Therefore by the IVT there exists a number c in the interval $(1, 2)$ such that $f(c) = 0$.

Therefore there is a number c such that $c^3 - 2c - 3 = 0$ and so the equation $x^3 - 2x - 3 = 0$ has a solution.

2. Show that $f(x) = \sqrt[3]{x} - 1 + x$ has at least one real root.

f is continuous because f is the sum of a polynomial and a radical function.

Note that:

$$f(0) = \sqrt[3]{0} - 1 + 0 = -1 < 0$$

$$f(1) = \sqrt[3]{1} - 1 + 1 = 1 > 0$$

Therefore by the Intermediate Value Theorem, there exists a number c in the interval $(0, 1)$ such that $f(c) = 0$. Therefore we know that f has a real root in the interval $(0, 1)$.

3. Show that the equation $\cos(x) = x^3$ has at least one real solution.

We rewrite the equation: $\cos(x) - x^3 = 0$.

Let $f(x) = \cos(x) - x^3$. We want to know if $f(x)$ has a root.

f is continuous because f is the difference of a trigonometric function (continuous everywhere) and a polynomial.

Note that:

$$f(0) = \cos(0) - 0^3 = 1 - 0 = 1 > 0$$

$$f(\pi) = \cos(\pi) - \pi^3 = -1 - \pi^3 < 0$$

Therefore by the Intermediate Value Theorem, there exists a number c in the interval $(0, \pi)$ such that $f(c) = 0$. Therefore we know that $\cos(x) = x^3$ has a solution in the interval $(0, \pi)$.

4. Show that the equation $\ln(x) = 3 - 2x$ has at least one real solution.

We rewrite the equation: $\ln(x) - 3 + 2x = 0$.

Let $f(x) = \ln(x) - 3 + 2x$. We want to know if $f(x)$ has a root.

f is continuous because f is the sum of a logarithmic function and a polynomial.

Note that:

$$f(1) = \ln(1) - 3 + 2(1) = 0 - 3 + 2 = -1 < 0$$

$$f(e) = \ln(e) - 3 + 2e = -2 + 2e > 0 \quad (\text{since } e \approx 2.72)$$

Therefore by the Intermediate Value Theorem, there exists a number c in the interval $(1, e)$ such that $f(c) = 0$. Therefore we know that $\ln(x) = 3 - 2x$ has a solution in the interval $(1, e)$.

5. Is there a number that is exactly 1 more than its cube? Use the IVT to answer this question.

We are interested in finding a solution to the equation: $x = x^3 + 1$.

We rewrite the equation: $x - x^3 - 1 = 0$

Let $f(x) = x - x^3 - 1$. We want to know if $f(x)$ has a root.

f is continuous because f is a polynomial.

Note that:

$$f(-1) = (-1) - (-1)^3 - 1 = -1 + 1 - 1 = -1 < 0$$

$$f(-2) = (-2) - (-2)^3 - 1 = -2 + 8 - 1 = 5 > 0$$

Therefore by the Intermediate Value Theorem, there exists a number c in the interval $(-2, -1)$ such that $f(c) = 0$. Therefore we know that $x = x^3 + 1$ has a solution in the interval $(-2, -1)$. So there does exist a number that is exactly 1 more than its cube.